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English version

Digital product passport - Data storage, archiving, and data persistence

Passeport numérique des produits - Stockage, archivage et persistance des données

Digitaler Produktpass - Datenspeicherung, Archivierung und Datenpersistenz

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Contents

Page

European foreword ... 3

Introduction ... 4

1 Scope ... 5

2 Normative references ... 5

3 Terms and definitions ... 5

4 General principles and requirements ... 6
4.1 Requirements for storage of the digital product passport data ... 6
4.2 Archiving and rules for archiving ... 7
4.3 Data persistence and rules for data persistence ... 7
4.4 Additional digital documentation attached to digital product passport ... 7
4.5 Replication between economic operators and back-up digital product passport service providers ... 8

Annex ZA (informative) Relationship between this European Standard and the essential requirements of (EU) Regulation No 2024/1781 aimed to be covered ... 9

Bibliography ... 10

European foreword

This document (EN 18221:2026) has been prepared by Technical Committee CEN/CLC/JTC 24 “Digital Product Passport – Framework and System”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by November 2026, and conflicting national standards shall be withdrawn at the latest by November 2026.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

This document has been prepared under a standardization request addressed to CEN by the European Commission. The Standing Committee of the EFTA States subsequently approves these requests for its Member States.

For the relationship with EU Legislation, see informative Annex ZA, which is an integral part of this document.

Any feedback and questions on this document should be directed to the users’ national standards body. A complete listing of these bodies can be found on the CEN website.

According to the CEN-CENELEC Internal Regulations, the national standards organisations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Republic of North Macedonia, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Türkiye and the United Kingdom.

Introduction

A digital product passport (DPP) is a key enabling mechanism to make product information traceable and accessible across value chains - supporting economic operators, manufacturers, distributors, repairers, recyclers and consumers to make informed decisions and to support a circular economy. The implementation of digital product passports will be carried out progressively. Sector-specific initiatives will determine the precise DPP content and requirements for individual product groups.

To support the implementation of DPPs, 8 standards have been developed so far:

— EN 18219:2026 – Digital product passport – Unique identifiers

— EN 18220:2026 – Digital product passport – Data carriers

— EN 18216:2026 – Digital product passport – Data exchange protocols

— EN 18222:2026 – Digital Product Passport – Application Programming Interfaces (APIs) for the product passport lifecycle management and searchability

— EN 18223:2026 – Digital Product Passport – System interoperability

— EN 18221:2026 – Digital product passport – data storage, archiving, and data persistence (this document)

— EN 18239:—[1] – Digital Product Passport – access rights management, information system security, and business confidentiality

— EN 18246:—[2] – Digital Product Passport – Data authentication, reliability and integrity

[1] Under preparation. Stage at the time of publication: prEN 18239:2025.
[2] Under preparation. Stage at the time of publication: prEN 18246:2025.

1 Scope

This document specifies requirements for data storage, archiving, and data persistence of digital product passports, all based on a decentralized approach. The archiving functionality securely stores historical passport data, preserving a comprehensive record of past information. Persistence ensures that data included in the digital product passport remains available even when the economic operator creating the digital product passport is no longer active.

This document also specifies requirements for the replication between economic operators and back-up operators as well as rules for data lifetime definition.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 18216, *Digital product passport - Data exchange protocols*

EN 18220, *Digital product passport - Data carriers*

EN 18222, *Digital Product Passport - Application Programming Interfaces (APIs) for the product passport lifecycle management and searchability*

EN 18239:—[3], *Digital Product Passport - access rights management, information system security, and business confidentiality*

EN 18246:—[4], *Digital product passport - Data authentication, reliability and integrity*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at http://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org/

3.1
economic operator
manufacturer, authorized representative, importer, distributor, dealer or fulfilment service provider

3.2
digital product passport
DPP
digital record of product characteristics throughout its life cycle

Note 1 to entry: Example characteristics include environmental sustainability, environmental impact, and recyclability

[3] Under preparation. Stage at the time of publication: prEN 18239:2025.
[4] Under preparation. Stage at the time of publication: prEN 18246:2025.

3.3
digital product passport service provider
natural or legal person that is an independent third party authorized by the *economic operator* (3.1) that places the product on the market or puts it into service and that processes the digital product passport data for that product for the purpose of making such data available to economic operators and other relevant actors with a right to access those data

3.3.1
main digital product passport service provider
digital product passport service provider (3.3) that is hosting the *digital product passport* (3.2) on behalf of the *economic operator* (3.1) if the *economic operator* chooses not to host the digital product passport themselves

3.3.2
back-up digital product passport service provider
digital product passport service provider (3.3) that is hosting the back-up copy of the *digital product passport* (3.2)

3.4
digital product passport lifetime
period during which the *digital product passport* (3.2) is required by regulation to be available to any operator or natural or legal person with a right to access its data

4 General principles and requirements

4.1 Requirements for storage of the digital product passport data

Digital product passport shall be stored by the actor responsible for its creation, or by digital product passport service providers acting on their behalf.

The data in the digital product passport that are stored shall be accurate, complete and reflecting all relevant changes up to the current state.

The digital product passport shall be made accessible online via exchange protocols as defined in EN 18216 during the digital product passport lifetime.

A digital product passport may contain references to other digital product passports. Digital product passports are always stored independently of each other.

Digital product passport data shall be stored such that the required human-readable and machine-readable formats (as defined in EN 18216) can be generated from stored data.

The digital product passport architecture is decentralized, and this document does not impose specific data storage technologies. This allows economic operators and digital product passport service providers to leverage existing systems for the underlying storage.

Requirements for storage shall apply to both main and back-up service providers. Main and back-up service providers shall implement all measures based on proven reliable technologies to ensure data availability, integrity, authenticity, and protection against data loss, malicious modification, and unauthorized access or tampering, also in accordance with EN 18239[3] and EN 18246[2].

Interoperability specifications, data formats, and semantic data models used in storage are defined in EN 18223 [1].

Digital product passport data shall be stored in a way supporting the needs of EN 18220 and EN 18222.

4.2 Archiving and rules for archiving

The archiving functionality stores historical passport data, preserving a comprehensive record of past information.

Archiving starts when the first change of the initial digital product passport occurs.

Archived versions of the digital product passport shall be stored both by the back-up digital product passport service provider, and by the creator of the digital product passport or a main digital product passport service provider operating on its behalf.

All archived versions shall be maintained during the digital product passport lifetime.

All changes to the digital product passport shall be archived. However, changes explicitly identified as exempted by product-specific requirements (e.g. real-time sensor data) may be excluded.

Attributes of archived versions are subject to the same access restrictions as the corresponding attributes in the current digital product passport.

The archived version corresponding to a given point in time shall be retrievable by authenticated and authorized actors.

Integrity and originality of the archived versions shall be ensured, following EN 18246[2] on data authentication, reliability, integrity.

For archiving, digital product passport service providers and back-up digital product passport service providers should adhere to ISO 14721 [2] (OAIS): taking into account the subclauses 1.4 (Conformance), 2.3 (OAIS information), 2.4 (preservation objectives), 3.2 (mandatory responsibilities), 4.2.2 (general model), 4.3 (information model). Implementations should also consider Clause 6 (archiving interoperability) for guidance.

4.3 Data persistence and rules for data persistence

When a product is placed on the market, it can be required to make available a back-up copy of the digital product passport through a digital product passport service provider which shall store a back-up copy of the most up-to-date version and all required historical versions of the digital product passport.

The back-up digital passport attributes are subject to the same access restrictions as the corresponding attributes in the original digital product passport.

The digital product passport shall be available to specifically authorized users from the back-up digital passport service provider during the digital product passport lifetime.

General access to the digital product passport from the back-up digital product passport service provider shall be available once the economic operator is no longer active on the market.

4.4 Additional digital documentation attached to digital product passport

When additional digital documentation is provided, it shall be included, directly or by reference, in the digital product passport and made accessible through the digital product passport.

The digital documentation shall be presented in a format that makes it possible to download and save it on an electronic device so that the user can access it at all times and shall be made accessible online during the digital product passport lifetime.

The back-up digital product passport service provider shall be responsible for storing and making accessible mandatory digital documentation included whether directly or by reference in the digital product passport.

4.5 Replication between economic operators and back-up digital product passport service providers

The replication of the digital product passport and all changes shall be ensured as required by 4.3 to its back-up digital product passport service provider.

The frequency of this replication in the case of modifications, shall be sufficient to minimize data loss (e.g. defining a maximum Recovery Point Objective [RPO]) to safeguard the data in the case of an economic operator leaving the market, but the frequency may be further defined by any requirements specific to the product.

Replication between the economic operator and its back-up digital product passport service provider shall be possible through the use of a standardized digital product passport lifecycle API according to EN 18222 but may also be performed with other mutually agreed secure replication mechanisms.

Replication between the economic operator and its back-up digital product passport service provider shall take place over a data exchange protocol in accordance with EN 18216.

Annex ZA
(informative)

Relationship between this European Standard and the essential requirements of (EU) Regulation No 2024/1781 aimed to be covered

This European Standard has been prepared under a Commission's standardization request M/604 Amd 1 to provide one voluntary means of conforming to essential requirements of (EU) Regulation 2024/1781 of 13 June 2024 of the European Parliament and of the Council of the European Union establishing a framework for the setting of ecodesign requirements for sustainable products, amending Directive (EU) 2020/1828 and Regulation (EU) 2023/1542 and repealing Directive 2009/125/EC.

Once this standard is cited in the Official Journal of the European Union under that Regulation, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding essential requirements of that Regulation, and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and Articles of (EU) Regulation 2024/1781

Essential requirements of [EU] Regulation 2024/1781	Clause(s)/subclause(s) of this EN	Remarks/Notes
Article 10.4	4.3, 4.5	presumption of conformity only for the technical aspects of the backup not for the existence of the backup as such
Article 11.c	4.5	

WARNING 1 — Presumption of conformity stays valid only as long as a reference to this European Standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING 2 — Other Union legislation may be applicable to the products falling within the scope of this standard.

Bibliography

[1] EN 18223, *Digital Product Passport - System interoperability*

[2] ISO 14721:2025, *Space Data System Practices — Reference model for an open archival information system (OAIS)*

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